A Project Report on

PG Management System

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

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Abstract

The PG Management System is a web-based application designed to streamline and digitize the management of Paying Guest (PG) accommodations. With increasing urbanization, the demand for organized and efficient housing solutions has significantly grown. However, many PG systems still rely on manual processes, leading to inefficiencies, data loss, and lack of transparency between administrators and tenants. This project aims to develop a centralized platform that enables efficient tenant management, complaint tracking, room allocation, and life cycle monitoring. The system provides role-based access for administrators and tenants, ensuring secure and transparent interactions. Advanced features such as encrypted password storage using bcrypt, real-time complaint handling, and automated life cycle tracking improve both usability and security. The system is developed using modern web technologies including Node.js, Express.js, MySQL, and front end technologies such as HTML, CSS, and JavaScript. The implementation demonstrates improved performance, reduced manual errors, and faster data processing compared to traditional systems.

Overall, the PG Management System contributes to Sustainable Development Goal 11 (Sustainable Cities and Communities) by promoting organized urban living through digital infrastructure. Future enhancements may include payment integration, mobile application support, and multi-property scalability.

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ABBREVATION

AI	*Artificial Intelligence*
ML	*Machine Learning*
PG	*Paying Guest*
SMS	*Short Message Service*
CRUD	*Create, Read, Update and Delete*
API	*Application Programming Interface*
SaaS	*Software as a Service*
ER	*Entity Relationship*
PGMS	*Paying Guest Management System*
ERP	*Enterprise Resource Planning*
HTML	*Hyper Text Markup Language*
CSS	*Cascading Style Sheets*
SQL	*Structured Query Language*
STEM	*Science, Technology, Engineering and Mathematics*
ISSN	*International Standard Serial Number*
IEEE	*Institute of Electrical and Electronics Engineers*

Chapter 1

Introduction

1.Introduction

1.1 Background & Importance

Urbanization has led to a rapid migration of people from rural and semi-urban areas toward cities in search of better education, employment, and living opportunities. This growing population in urban areas has significantly increased the demand for affordable and flexible housing solutions such as Paying Guest (PG) accommodations. PGs provide a convenient and cost-effective alternative to renting apartments, especially for students and working professionals who require short-term or shared living arrangements. Despite this rapid growth, many PG accommodations still rely on traditional manual methods for managing their operations. These methods include maintaining physical registers for tenant records, rent payments, and complaints, which are not only time-consuming but also highly prone to errors, data loss, and mismanagement. As the number of tenants increases, handling such records manually becomes increasingly difficult and inefficient. In today's digital era, there is a strong need for a modern, technology-driven solution that can streamline PG management processes. A digital system can help in organizing data, improving communication, and ensuring data security. The proposed PG Management System addresses these needs by providing a centralized platform that enables efficient handling of all PG-related activities.

Furthermore, the system contributes to Sustainable Development Goal 11 (Sustainable Cities and Communities) by promoting organized, efficient, and technology-enabled urban living. By encouraging the adoption of digital infrastructure, the system helps in improving the quality of urban services, reducing inefficiencies, and supporting sustainable development in growing cities.

1.2 Current Challenges

The existing PG management system faces several challenges that significantly impact its efficiency, reliability, and overall usability. One of the primary issues is the manual

management of tenant records, where all information is written and maintained by hand. This process is not only time-consuming but also increases the likelihood of human errors, such as incorrect entries, duplication of records, or loss of important data. Another major challenge is the absence of a centralized complaint tracking system. In most cases, communication between tenants and administrators is done verbally or through informal messaging, which can lead to loss of important details, misunderstandings, or complete neglect of complaints. Without a proper system to record and track complaints, it becomes difficult to ensure accountability and timely resolution of issues.

The lack of life cycle tracking of tenants is also a significant problem. There is no structured way to track whether a tenant is currently active, has given notice, or has already vacated the premises. This makes it difficult for administrators to accurately manage room occupancy and plan future allocations, often resulting in inefficient use of available space and potential financial losses. Additionally, there is a lack of transparency between tenants and administrators. Tenants do not have direct access to their own information, such as payment status, complaint updates, or duration of stay. This lack of visibility can lead to confusion, misunderstandings, and reduced trust between both parties, ultimately affecting the overall management experience.

1.3 Motivation

The current system faces several challenges that affect its efficiency and reliability. Manual tenant record management is a time-consuming process, as all details must be written and maintained by hand, increasing the chances of errors and data loss. Additionally, there is no centralized complaint tracking system, and reliance on manual communication often leads to loss of important details or forgetfulness. Another major issue is the absence of proper life cycle tracking of tenants (such as Active, Notice, or Left), making it difficult to accurately determine occupancy status, which can result in inefficient use of available PG accommodations and potential financial loss. Furthermore, there is a lack of transparency between tenants and administrators, which can lead to misunderstandings and reduced trust between both parties.

Chapter 2

2. Literature Survey

Existing PG management systems provide a range of basic functionalities that assist in managing daily operations of paying guest accommodations. These systems primarily focus on maintaining tenant records, where essential details such as tenant names, contact information, and joining dates can be stored and accessed. They also support room or bed allocation, allowing administrators to assign available rooms to tenants through a web-based interface. In addition, rent tracking and billing features are commonly included, ensuring that payment details are recorded, monitored, and maintained systematically.

Many systems also incorporate communication features such as payment reminders sent via SMS or WhatsApp, helping tenants stay informed about due payments. A basic reporting dashboard is often available, which provides a centralized view of alerts, occupancy status, and other key information. These features collectively aim to simplify management tasks and reduce dependency on manual record-keeping to some extent.

2.1 Analysis of the weaknesses/limitations presented in surveyed solutions

Despite offering essential functionalities, existing PG management systems suffer from several limitations that reduce their effectiveness. One of the major issues is the owner-centric design, where most control and visibility are restricted to the administrator. This lack of transparency prevents tenants from accessing their own information, such as payment status, complaint updates, or remaining stay duration, leading to dependency and reduced trust. Another significant limitation is the poor user interface and user experience (UI/UX). Many systems are not designed with usability in mind, making it difficult for users to navigate or manage data efficiently. The absence of an intuitive and well-structured interface often results in confusion and reduced adoption.

Communication between administrators and tenants is also limited, as most systems do not provide a proper platform for interaction or feedback. This leads to delays in addressing issues and creates a gap in coordination. Furthermore, tenants face several restrictions, as many systems do not allow them to directly file complaints, track their status, or request additional services from within their dashboard. A critical drawback is the lack of automation in handling tasks such as complaint management and notifications. Complaints are often ignored, misplaced, or not properly tracked, and there is no reliable mechanism for tenants to follow up

or provide proof. These shortcomings highlight the need for a more advanced, transparent, and user-centric system.

2.2 Survey of the methodologies/techniques used in the existing solutions

Existing PG management systems are built using a combination of standard software development methodologies and technologies. One of the most widely used approaches is CRUD-based data management, which enables basic operations such as creating, reading, updating, and deleting tenant and rent records. This forms the foundation of most systems and ensures structured data handling through relational databases. These systems typically follow a client-server architecture, where the front end is responsible for user interaction and the back end handles data processing and business logic. Communication between these layers is managed through APIs, ensuring efficient data exchange between users and the database.

In recent years, many systems have adopted a cloud-based Software as a Service (SaaS) model. This approach allows users to access the system from any location using internet-enabled devices, while also providing benefits such as scalability, data backup, and reduced infrastructure costs. It also enables multi-user access and supports the growing needs of modern applications. Another important technique used in these systems is third-party API integration. Services such as payment gateways, SMS notifications, and messaging platforms like WhatsApp are integrated to provide additional functionalities, including online payments and automated communication with tenants. Furthermore, basic data analytics techniques are applied to generate reports such as monthly income, occupancy rates, and payment summaries. These insights help administrators in decision-making and monitoring system performance. However, most existing systems are limited to simple analytical capabilities and do not incorporate advanced or predictive analytics, which could further enhance operational efficiency.

Table 1: Literature Survey

Author / Source	Technique Used	Platform	Key Results	Limitations
Zapstel	Cloud-based SaaS, CRUD operations, Rule-based automation	Web+Mobile	Efficient tenant management, automated rent tracking, improved communication	Limited tenant-centric features, basic analytics
MyPGMS	Database management, Dashboard analytics, API integration	Web	Centralized data handling, improved reporting and tracking	Complex UI, lacks advanced automation
PG Master	Mobile-based system, Real-time data updates	Mobile App	Easy multi-property management, real-time occupancy tracking	Limited scalability, minimal analytics
StayLeap	ERP-based architecture, Cloud computing	Web+Mobile	End-to-end PG/hostel management, improved operational efficiency	High complexity, less user-friendly for small PG owners

Chapter 3

Limitation of Existing system

3. Limitation of Existing system

Existing systems suffer from several significant limitations that affect their usability, efficiency, and overall user satisfaction. One of the major issues is the owner-centric design, which results in a lack of transparency between administrators and tenants. In such systems, tenants are not provided with direct access to their own data, such as payment records, complaint status, or remaining duration of stay. This creates a strong dependency on the administrator even for basic information, which reduces convenience and can lead to mistrust among users. Another important limitation is the weak user interface and poor user experience (UI/UX). Many existing systems are not designed with user-friendliness in mind, leading to unstructured data presentation and complicated navigation. As a result, both administrators and tenants may find it difficult to efficiently manage or retrieve information. This lack of an intuitive interface reduces the overall effectiveness of the system and discourages consistent usage. Communication between administrators and tenants is also highly limited in traditional systems. There is often no dedicated platform for interaction, feedback, or updates, which leads to communication gaps. This can result in misunderstandings, delayed responses to issues, and dissatisfaction among tenants. Without proper communication channels, it becomes challenging to maintain smooth coordination between users.

Furthermore, tenants face several functional limitations. Many systems do not provide features for tenants to independently file complaints, track their status, or request additional services such as food or maintenance. This restricts tenant involvement and reduces their ability to actively participate in the management process. A critical drawback of existing systems is the lack of automation. Tasks such as complaint handling, notifications, and record updates are often performed manually or are not properly tracked. Complaints may be ignored, misplaced, or forgotten, and there is no reliable mechanism to maintain proof or history of requests.

Overall, these limitations highlight the need for a more advanced solution that focuses on transparency, user-friendliness, automation, and secure data handling. Addressing these issues is essential for improving the efficiency and reliability of PG management systems.

Chapter 4

4. Problem Statement and Objectives

Traditional PG management systems are largely admin-centric, providing limited transparency between administrators and tenants. In such systems, tenants have little or no access to their own information, including payment records, complaint status, and duration of stay. This lack of visibility often leads to mistrust, confusion, and a sense of insecurity among tenants. Additionally, most existing systems rely on manual processes or partially digitized methods, which are inefficient and prone to errors. The absence of a centralized system further complicates PG management, as data is often scattered across different records or maintained manually. This results in difficulties in tracking tenant information, managing room allocations, and handling complaints effectively. Communication between administrators and tenants is also limited, leading to delays in addressing issues such as maintenance requests, food services, and important notifications. Moreover, the lack of automation in existing systems means that complaints and requests are often ignored, misplaced, or not properly documented. This creates a lack of accountability and makes it difficult to ensure timely resolution of problems. Therefore, there is a need for a secure, automated, and web-based solution that can centralize operations, improve transparency, enhance communication, and streamline overall PG management processes.

The proposed system aims to centralize all PG operations into a single, unified platform, ensuring that data and processes are managed efficiently in one place. It implements role-based access control, allowing different functionalities for administrators and tenants, thereby enhancing security and usability. The system also enables effective complaint monitoring, ensuring that all issues raised by tenants are properly recorded, tracked, and resolved in a timely manner. Additionally, it provides a feature for administrators to send alerts and notifications, helping in better communication and ensuring that important information is conveyed to tenants promptly.

Chapter 5

5. Proposed System

The proposed PG Management System is a web-based application designed to provide a centralized, secure, and efficient platform for managing paying guest accommodations. It focuses on improving transparency, automating manual tasks, and enhancing communication between tenants and administrators through a structured digital system.

5.1 Analysis/Framework/Algorithm

5.1.1 System Analysis

The system is designed using a client-server architecture, where the front end (client) is responsible for handling user interaction and providing an interface for both administrators and tenants. The back end (server) manages the application logic and processes user requests while communicating with the database. The database stores all structured information, including details related to tenants, rooms, complaints, and payments. This architectural approach ensures better scalability, making it easier to expand the system in the future, while also improving maintainability and enabling efficient data processing.

5.1.2 Framework Used

The system is developed using a combination of modern web technologies to ensure efficiency, scalability, and security. The back end is built using Node.js, which serves as the runtime environment, along with Express.js as the web framework for handling APIs and managing server-side operations. MySQL is used as the relational database for storing structured data such as tenant details, room information, complaints, and payments. The front end interface is developed using HTML, CSS, and JavaScript to provide a user-friendly and interactive experience. Additionally, the bcrypt library is used for password hashing, ensuring secure authentication and protection of user credentials.

5.1.3 Algorithm

The system follows a structured workflow for handling user interactions and operations. During the user registration and login process, the user enters their credentials, and the password is securely hashed using bcrypt before being stored in the database. During login, the entered credentials are validated against the stored data to ensure secure access.

Once logged in, tenants can perform various operations such as viewing their personal details and tenure information, filing complaints which are stored in the database, and applying for a vacating notice when they plan to leave the PG.

Complaint Handling Algorithm:

Step 1: Tenant submits complaint

Step 2: Complaint stored with status = "Pending"

Step 3: Admin views complaint

Step 4: Admin updates status → "In Progress" / "Resolved"

Step 5: Updated status reflected in tenant dashboard

5.2 Design Details

The system is designed based on a three-tier architecture to ensure modularity, scalability, and efficient data handling. The first layer is the Presentation Layer (Front end), which provides the user interface for both administrators and tenants. It is developed using HTML, CSS, and JavaScript to ensure an interactive and user-friendly experience. The second layer is the Application Layer (Back end), which handles the core business logic of the system.

5.3 Methodology

The methodology of the proposed system focuses on digitizing and automating PG management using modern web technologies. The process begins with requirement analysis, where the issues in traditional PG systems are identified, user roles such as Admin and Tenant are defined, and the necessary functional requirements are gathered. This is followed by system design, which involves designing the database schema, planning the system architecture, and defining API endpoints for communication between components.

In the development phase, the front end is built using HTML, CSS, and JavaScript, while the back end is developed using Node.js and Express.js. These components are then integrated with a MySQL database to enable efficient data storage and retrieval. Security implementation is an important step, where bcrypt is used for password hashing to ensure secure authentication, along with measures to prevent unauthorized access and maintain data privacy.

After development, the system undergoes testing, including unit testing for individual modules, integration testing to ensure all components work together, and performance testing

to evaluate system efficiency. This structured approach ensures that the final system is reliable, secure, and efficient.

5.4 Proposed System Features

The proposed system includes several features that improve the overall functionality and usability of PG management. It provides centralized data management, allowing all information to be stored and accessed from a single platform. A secure login system ensures that user credentials are protected, while the complaint tracking feature allows tenants to raise issues and monitor their status. The system also includes tenant life cycle management, enabling tracking of tenant status throughout their stay. Efficient room allocation helps in better utilization of available space, and an improved communication system enhances interaction between administrators and tenants.

5.5 Advantages of Proposed System

The proposed system offers several advantages over traditional methods. It significantly reduces manual errors by automating data handling and record management. It saves time and effort for both administrators and tenants by simplifying routine tasks. The system improves transparency by allowing users to access relevant information easily.

It also enhances the overall user experience through a structured and user-friendly interface. Additionally, secure data handling ensures that sensitive information is protected, making the system reliable and trustworthy.

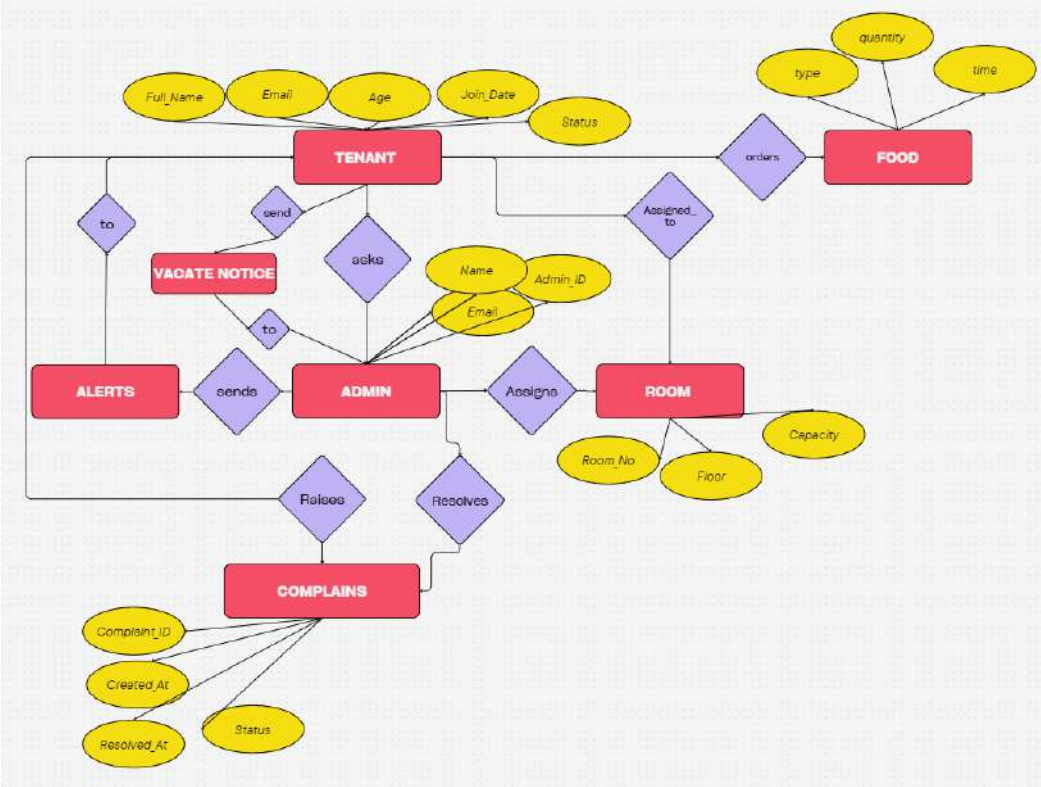

Figure 1:ER Diagrams

This ER diagram represents a Hostel/PG Management System that revolves around two core entities: Tenant and Admin. A Tenant has attributes such as Full_Name, Email, Age, Join_Date, and Status, while an Admin is identified by their Name, Admin_ID, and Email.

At the heart of the system, the Admin assigns Rooms to Tenants, where each Room is characterized by its Room_No, Floor, and Capacity. Tenants can also place food orders, with the Food entity tracking the type, quantity, and time of each order. When a Tenant wishes to leave, they can submit a Vacate Notice to the Admin, and if they have any queries, they can directly approach the Admin as represented by the "asks" relationship.

The system also handles communication and issue management. Admins can send Alerts to Tenants to broadcast important information. If any issues arise, Admins can raise Complaints, which are then tracked under the Complains entity with attributes like Complaint_ID, Created_At, Resolved_At, and Status. The Admin is also responsible for resolving these complaints, closing the feedback loop. Overall, this diagram captures the complete life cycle of tenant management — from room assignment and food ordering to complaint resolution and vacate processing.

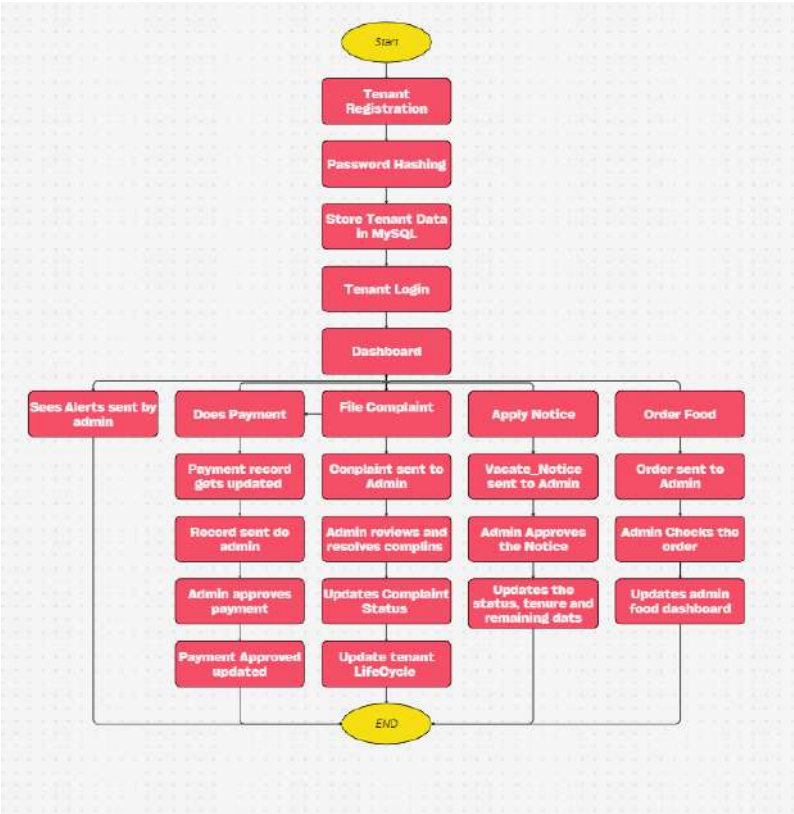

Figure 2:Flow Diagram

The system begins with Tenant Registration, where the tenant's password is securely hashed and their data is stored in a MySQL database. Once registered, the tenant can log in and access their personal Dashboard, which serves as the central hub for all available actions.

From the dashboard, the tenant has five key functionalities. First, they can view Alerts sent by the Admin, keeping them informed of any important notices. Second, they can make Payments, where the payment record gets updated, sent to the Admin, and upon Admin approval, the payment status is marked as approved. Third, tenants can File a Complaint, which is sent to the Admin for review. The Admin then resolves the complaint, updates the complaint status, and finally updates the tenant's life cycle accordingly. Fourth, tenants can Apply for a Vacate Notice, which triggers a vacate notice to be sent to the Admin. The Admin reviews and approves the notice, following which the tenant's status, tenure, and remaining dates are updated in the system. Fifth and finally, tenants can Order Food, where the order is sent to the Admin, who checks and verifies it, and the food dashboard on the Admin's side is updated to reflect the new order. Overall, this flowchart captures the complete operational workflow of the system, from onboarding a tenant to managing their day-to-day activities through an admin-supervised process.

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Dataset details

- Tenants (ID, Name, Contact, Room, Join Date)
- Admins (ID, Username, Password)
- Rooms (Room ID, Capacity, Availability)
- Complaints (Complaint ID, Tenant ID, Status, Description)
- Payments (Payment ID, Amount, Date, Status)
- Vacate Notices (Tenant ID, Notice Date, Status)

6.2 Performance Evaluation Paramters

To evaluate the effectiveness and efficiency of the proposed PG Management System, several system-level performance metrics are considered. These parameters help measure improvements over traditional manual systems.

1. Response Time

Measures the time taken by the system to respond to user requests. Includes operations like login, data retrieval, complaint submission, and updates.

Expected Result: Less than 2 seconds for most operations.

Importance: Ensures smooth user experience and system responsiveness.

2. Accuracy

Refers to correctness of data processing such as rent calculation, tenant records, and complaint status.

Expected Result: Around 95–98% accuracy.

Importance: Reduces errors compared to manual record-keeping.

3. Error Rate Reduction

Measures decrease in manual errors (duplicate entries, incorrect records).

Expected Result: ~80–85% reduction compared to traditional systems.

Importance: Improves reliability of the system.

4. Task completion Time:

Time required to perform operations like adding a tenant, updating records or filing a complaint.

Expected Result: 50–60% faster than manual methods.

Importance: Enhances operational efficiency.

5. System Throughput

Number of requests handled by the system per unit time. Example: Number of users accessing the system simultaneously.

Expected Result: Efficient handling of multiple users without delay.

Importance: Indicates scalability of the system.

6. Data Security Level

Evaluates how well user data is protected.

Expected Result: High security with encrypted credentials.

Importance: Prevents unauthorized access and data breaches.

7. Availability

Measures system uptime and accessibility.

Expected Result: 24/7 availability with minimal downtime.

Importance: Ensures users can access the system anytime.

6.3 Software/Hardware Setup

- Node.js : Used as a connection to all the back end database and the front end. Using APIs
- Express.js: Assistance the Node.js with the APIs
- MySQL: Used as a database for storing the data.
- HTML / CSS / JavaScript: For our frontend website and for our backend.
- bcrypt library: Hashing of the passwords for privacy.

Chapter 7

Results and Discussion

7. Results and Discussion

The implementation of the PG Management System demonstrates significant improvements in overall system performance, efficiency, and reliability when compared to traditional manual methods. One of the key improvements observed is in the accuracy of operations, as the system ensures correct rent calculations and proper maintenance of tenant records through automated data handling. By eliminating manual calculations and handwritten entries, the chances of inconsistencies and errors are greatly reduced, resulting in more reliable and trustworthy data management. Another important improvement is the reduction in response time, which allows users to perform operations such as logging in, retrieving data, updating records, and submitting complaints quickly and efficiently. This faster interaction enhances the overall user experience and ensures that both administrators and tenants can access the required information without unnecessary delays. Additionally, the system significantly reduces manual errors that are common in register-based systems, such as duplicate entries, missing records, and incorrect data updates. The digital approach ensures better data validation and consistency, minimizing such issues.

Furthermore, the system has enhanced data management efficiency by storing all information in a centralized database. This centralization ensures that data is easily accessible, well-organized, and can be retrieved whenever required. It also simplifies the process of updating and maintaining records, reducing redundancy and improving overall system organization. This increased visibility helps in building trust and reducing misunderstandings. When compared to traditional systems, the proposed solution offers several advantages, including faster processing of tasks, reduced errors due to automation, and improved communication between users. The system provides a more organized and user-friendly platform that simplifies complex management processes. Overall, the PG Management System demonstrates improved performance, better reliability, enhanced usability, and greater efficiency, making it a highly effective solution for modern PG management needs.

7.2 Data Table

Table 2:Time Efficiency Data Table

Tenant No.	Manual System Time (min)	Proposed System Time (min)
1	20	8
2	22	9
3	18	7
4	25	10
5	21	8
6	23	9
7	19	7
8	24	10

From above data, it can be concluded that proposed methods are about 60% faster than surveyed methods.

Table 3:Collection Data Table

Month	Total Tenants	Tenants Paid	Efficiency (%)
Jan	50	25	50
Feb	55	33	60
Mar	60	39	65
Apr	65	49	75
May	70	60	86
Jun	75	71	95

7.3 Graphical Representation

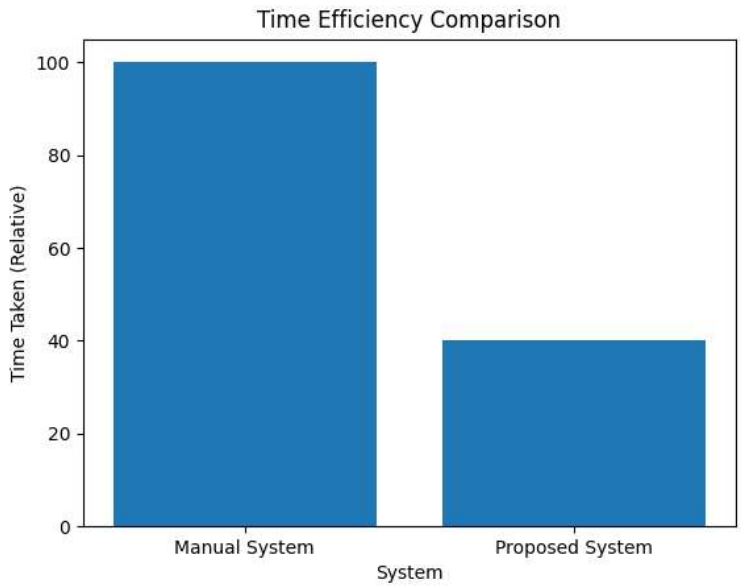

Figure 3:Time Efficiency Chart

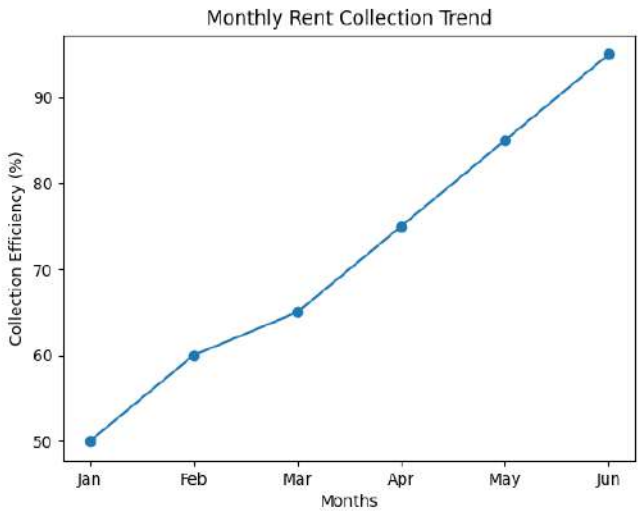

Figure 4:Collection Comparison Graph

7.4 Comparison with Existing Systems

Table 4:Existing Systems vs Proposed Systems

Feature	Existing Systems	Proposed System
Transparency	Low	High
Tenant Interaction	Limited	Full Access
Automation	Partial	High

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

The proposed PG Management System has been successfully developed as a comprehensive web-based application that effectively addresses the limitations of traditional PG management methods. By digitizing and automating routine tasks such as tenant record management, complaint handling, and room allocation, the system significantly improves overall operational efficiency and reduces the dependency on manual processes. This not only saves time and effort but also minimizes the chances of human errors and data inconsistencies. The system enhances transparency between administrators and tenants by providing both parties with access to relevant information, such as payment details, complaint status, and tenancy duration, thereby fostering trust and better communication. Furthermore, the project contributes to Sustainable Development Goal 11 by promoting organized, efficient, and technology-driven urban living solutions. The use of modern web technologies ensures that the system is both secure and scalable, with features such as encrypted authentication mechanisms protecting sensitive user data. The modular architecture of the system allows for easy maintenance and future upgrades, making it adaptable to evolving requirements. Overall, the system successfully meets its objectives of improving efficiency, transparency, and reliability in PG management.

Although the system provides a strong foundation for managing PG operations, there are several opportunities for further enhancement. One important improvement is the integration of an online payment system, which would enable tenants to make secure and convenient digital transactions directly through the platform. The addition of SMS and email alert systems can further strengthen communication by providing real-time notifications for rent reminders, complaint updates, and important announcements.

Moreover, the system can be extended to support multiple PG properties, allowing administrators to manage several accommodations through a single platform, thereby increasing its scalability and applicability in real-world scenarios. Another significant enhancement would be the development of a mobile application version of the system, which would improve accessibility and provide users with the convenience of managing their activities on the go.

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